75. The average rainfall in a city for the first four days was recorded to be 0.40 inch. The rainfall on the last two days was in the ratio of 4 : 3. The average of six days was 0.50 inch. What was the rainfall on the fifth day?

(a) 0.60 inch (b) 0.70 inch
(c) 0.80 inch (d) 0.90 inch

Ans. (c) :
The total rainfall for the first
(c) Four days = $4 \times 0.40 = 1.60$
the total rainfall for 6 days = 6×0.50
$= 3.00$
$\therefore$ The total rainfall for 2 days = $3.00 - 1.60$
$=1.40$
According to question,
$4x + 3x = 1.40$
$7x = 1.40$
$x = 0.20$
$\therefore$ The rainfall of 5th day = 4×0.20
$= 0.80$ inch

76. A 2-digit number is reversed. The larger of the two numbers is divided by It smaller one. What is the largest possible remainder?

(a) 9 (b) 27 (c) 36 (d) 45

Ans. (d) :
If two digit number = 49
then two digit reversed number = 94
when the larger number is divided by small number,
$\therefore \frac{94}{49} \Rightarrow$ Remainder = 45

77. The monthly incomes of X and Y are in the ratio of 4 : 3 and their monthly expenses are in the ratio of 3: 2. However, each saves Rs. 6,000 per month. What is their total monthly income?

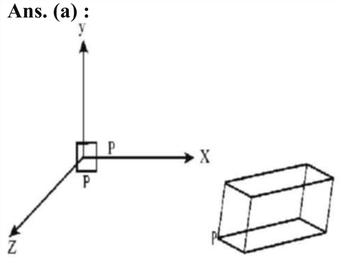
(a) Rs. 28,000 (b) Rs. 42,000
(c) Rs. 56,000 (d) Rs. 84,000

Ans. (b) :
Let the monthly income of x and y are 4k, 3k
According to question,
$\frac{4k-6000}{3k-6000} = \frac{3}{2}$ [$\because$ income – saving = Expenditure]
$2(4k - 6000) = 3(3k - 6000)$
$8k - 12000 = 9k - 18000$
$\boxed{6000 = k}$
$\because$ Total income x and y = $(4k + 3k)$
$= 7k$
$= 7 \times 6000$
$= 42000$

78. Two walls and a ceiling of a room meet at right angles at a point P. A fly is in the air 1 m from one wall, 8 m from the other wall and 9 m from the point P. How many meters is the fly from the ceiling?

(a) 4 (b) 6 (c) 12 (d) 15

Ans. (a) :



$\because$ $d = \sqrt{l^2 + b^2 + h^2}$
$l = 1$, b=8, h= ?
$9 = \sqrt{(1)^2 + (8)^2 + h^2}$
$9 = \sqrt{1 + 64 + h^2}$
$9 = \sqrt{65 + h^2}$
$81 = 65 + h^2$
$h^2 = 81 - 65$
$h^2 = 16$
$\boxed{h = 4}$

79. There are certain 2-digit numbers. The difference between the number and the one obtained on reversing it is always 27. How many such maximum 2-digit numbers are there?

(a) 3 (b) 4
(c) 5 (d) None of the above

Ans. (d) :
Total 2-digit numbers satisfying the conditions given in the questions :
14, 25, 36, 47, 58, 69
According to question,
41–14 = 27 52–25=27
63–36=27 74–47 =27
85–50=27 96–69 =27
Hence, total number of required two-digit number=6

80. What is the total number of digits printed, if a book containing 150 pages is to numbered from 1 to 150?

(a) 262 (b) 342 (c) 360 (d) 450

Ans. (b) :

Page No.		Number of digit
From 1 to 9	→	$1 \times 9 = 9$
From 10 to 99	→	$2 \times 90 = 180$
From 100 to 150	→	$3 \times 51 = 153$

Total Numbers of digit printed = 342

UNION PUBLIC SERVICE COMMISSION

Civil Services (Preliminary Exam) - 2018

CSAT : PAPER-II

(Chapterwise Analysis with Explanation)

Time : 2 hours **Maximum Number : 200**

COMPREHENSION

Directions for the following 4 (four) items :
Read the following **four passages** and answer the items that follow. Your answers to these items should be based on the passages only.

Passage–1

Global population was around 1.6 billion in 1990–today it is around 7.2 billion and growing. Recent estimates on population growth predict a global population of 9.6 billion in 2050 and 10.9 billion in 2100. Unlike Europe and North America, where only three to four per cent of population is engaged in agriculture, around 47 per cent of India's population is dependent upon agriculture. Even if India continues to do well in the service sector and the manufacturing sector picks up, it is expected that around 2030 when India overtakes China as the world's most populous country, nearly 42 per cent of India's population will still be predominantly dependent on agriculture.

1. Which of the following is the most logical and rational inference that can be made from the above passage?

(a) Prosperity of agriculture sector is of critical importance to India.
(b) Indian economy greatly depends on its agriculture.
(c) India should take strict measures to control its rapid population growth.
(d) India's farming communities should switch over to other occupations to improve their economic conditions.

Ans. (a) The most logical and rational inference that can be made from the above passage, the prosperity of the agricultural sector is of critical importance to India because agriculture is the backbone of the Indian economy because about 47% of India's population is dependent on agriculture. Unlike India, not only in the major continents of the world like Europe and North America, only 3-4 percent of the population is engaged in agriculture.

Passage–2

Many pathogens that cause food borne illnesses are unknown. Food contamination can occur at any stage from farm to plate. Since most cases of food poisoning go unreported, the true extent of global food borne illnesses is unknown. Improvements in international monitoring have led to greater public awareness, yet the rapid globalization of food production increases consumers' vulnerability by making food harder to regulate and trace. "We have the world on our plates", says an official of WHO.

2. Which of the following is the most logical corollary to the above passage?

(a) With more options for food come more risks.
(b) Food processing is the source of all food borne illnesses.
(c) We should depend on locally produced food only.
(d) Globalization of food production should be curtailed.

Ans. (a) As the most logical and corollary from the above passage, there are more risks due to more choices of food as there has been an increase in public awareness due to the improvement in the international production, food is controlled and traced. Being more difficult, the vulnerability of the consumer has increased and the outbreak of diseases has also increased.

Passage–3

I am a scientist, privileged to be somebody who tries to understand nature using the tools of science. But it is also clear that there are some really important questions that science cannot really answer, such as : Why is there something instead of nothing? Why are we here? In those domains, I have found that faith provides a better path to answers. I find it oddly anachronistic that in today's culture there seems to be a widespread presumption that scientific and spiritual views are incompatible.

3. Which of the following is the most logical and rational inference that can be made from the above passage?

(a) It is the faith and not science that can finally solve all the problems of mankind.
(b) Science and faith can be mutually complementary if their proper domains are understood.
(c) There are some very fundamental questions which cannot be answered by either science of faith.
(d) In today's culture, scientific views are given more importance than spiritual views.